Deleting an anchor from a paired string action

Human Compression project

Let $G = \langle h_0, \dots, h_{r-1} \rangle$ be transitive on $2r+1$ points. The marks are nonidentity involutions satisfying the string relations. An edge of colour i is eligible if its endpoints belong to different orbits of $G_i = \langle h_j : j \neq i \rangle$. Assume that each colour has at least two eligible edges. A first anchor is moved by h_0 and fixed by all later marks; a last anchor is defined in the reversed order.

We acquire the complete generic paired-cut, equality-connectedness, square-neighbour and ladder proofs from the L044 recovery. In particular, in degree $2r > 8$ their actual equality ladder has no first or last anchor: every point moved by an end mark is moved by another mark. We also acquire the ambient interval deficit. If an intransitive contiguous subfamily has s marks, f fixed points and c nontrivial orbits, then

$$2s \leq 2r + 1 - f - c. \tag{1}$$

These statements use neither parity nor the intersection property.

Lemma 1. *There is at most one first anchor. If it exists and is b, then G_0 has exactly three orbits: $\{b\}$ and two nontrivial orbits.*

Proof. Suppose two first anchors b, w exist. For $H = G_0$, (1) gives $f + c \leq 3$, while $f \geq 2$ and $c \geq 1$. Thus $f = 2, c = 1$. Let $x = b^{h_0}$ and $z = w^{h_0}$. The involution cannot pair b, w, since that pair would be invariant under every generator. Consequently x, z are distinct points of the unique nontrivial H-orbit V, whose size is $2r - 1$. Every h_i, $i \geq 2$, fixes x, z by commutation with h_0. Transitivity on V makes h_1 move both.

The action of H on V is faithful, since H fixes b, w. Every mark remains nonidentity. Each original eligible edge of a colour $i > 0$ lies in V and remains eligible for the smaller omission subgroup of H. We have therefore obtained a paired string action of degree $2(r-1)+1$, rank $r-1$, and two first anchors. Induction is impossible at rank one, where an involution cannot be transitive on three points. This proves the first assertion.

Now let b be a first anchor. The same interval bound gives $f + c \leq 3$ for G_0, with $f, c \geq 1$. If $f = c = 1$, the only eligible h_0-edge is its edge at b, contrary to the paired hypothesis. The possibility $f = 2, c = 1$ would give a second first anchor: any further G_0 fixed point must be moved by h_0, by full transitivity. It has just been excluded. Hence $f = 1, c = 2$. □

Lemma 2. *Suppose $r \geq 5$ and there are anchors at both ends. Each end colour has exactly two eligible edges.*

Proof. Write $h_0 = (b\ x)\eta$, where b is the first anchor. Delete b and replace h_0 by η. The remaining permutation graph is connected, since b was a pendant vertex. The string relations persist: for $i \geq 2$, commutation with h_0 and fixation of b give fixation of x and commutation with η. The latter mark is nonidentity because a second eligible h_0-edge existed.

Every omission orbit on the remaining points is unchanged. For an omission containing h_0, deleting its pendant vertex b does not change paths between other vertices. In the omission of h_0, that vertex was already isolated. Thus every later colour retains its eligible edges, and colour zero loses exactly the edge bx. The last anchor survives: it is fixed by h_0 and is therefore neither b nor x.

If colour zero retained at least two eligible edges, the modified degree-$2r$ action would satisfy all hypotheses of the acquired paired equality ladder. That ladder has no last anchor, a

contradiction. Hence exactly one eligible edge remains. Reversing the labels proves the other endpoint assertion. No intersection property is assumed for the modified action. □

Theorem 3 (The four-point head)**.** *Suppose $r \geq 5$ and there are distinct first and last anchors b, c. There are distinct points b, x, u, v, z such that, with $V = \Omega \setminus \{b, x, u, v\}$,*

$$h_0 = (b\ x)(u\ v), \qquad h_1 = (x\ u)(v\ z)\beta, \tag{2}$$

where β is supported in $V \setminus \{z\}$. The group $K = \langle h_2, \ldots, h_{r-1} \rangle$ fixes the four omitted points and is transitive on V. On V it is a paired string action of rank $r-2$ and degree $2(r-2)+1$, with first anchor z and last anchor c. If the original marking has the intersection property, this actual restricted marking does too.

Proof. By Lemma 1, G_0 has orbits $\{b\}, A, B$; label them so $x = b^{h_0} \in A$. By Lemma 2, there is exactly one other eligible h_0-edge. Full transitivity forces it to join $u \in A$ to $v \in B$. The four points are distinct. Every h_i, $i \geq 2$, fixes b, x. It transports the unique eligible edge between A, B to itself and preserves each of those orbits, so it also fixes u, v.

Apply (1) to K. It gives $f + c \leq 5$. Since K is nontrivial and fixes the four displayed points, those are exactly its fixed points and the remaining V is one nontrivial orbit. If V lay in A, then $B = \{v\}$, contradicting Lemma 1. Consequently

$$A = \{x, u\}, \qquad B = \{v\} \cup V.$$

The mark h_1 swaps x, u and pairs v with some $z \in V$. Thus it has the form in (2); initially $h_0 = (b\ x)(u\ v)\eta$ with η supported in V. This η centralizes the transitive K-action. The last anchor c is fixed by h_0, h_1, so it belongs to V and is fixed by η. A centralizer of a transitive action that fixes a point is the identity. Thus $\eta = 1$.

For $i \geq 3$, fixation of v and commutation with h_1 imply fixation of z. Transitivity of K makes h_2 move z. The last anchor c remains a last anchor and differs from z, since h_1 moves z and fixes c. The tail marks fix the complement of V, and their original eligible edges remain eligible after passing to smaller omission groups. This proves the claimed paired action. Its faithful identification with the actual subgroup K also preserves every subgroup intersection. □

Lemma 4. *Under Theorem 3, G contains* $\mathrm{Alt}(\Omega)$.

Proof. In a nontrivial block system, every block has an odd size at least three. The mark h_0 moves only four points and therefore cannot exchange two blocks. All other marks fix b and preserve its block. This block is then G-invariant, contrary to transitivity. Thus G is primitive.

Choose $\sigma \in K$ moving z and put $g = h_1 \sigma h_1$. It fixes b, x, u and sends v to some $w \in B \setminus \{v\}$. The product of h_0 and its conjugate by g is a three-cycle on u, v, w (replacing g by its inverse if required by the conjugation convention).

For completeness, a primitive group containing a three-cycle contains the alternating group. Take the supports of all conjugates of that cycle. Their incidence components are invariant blocks, so primitivity makes their union connected and equal to Ω. Three-cycles on intersecting triples generate the alternating group on their union: intersection in two points gives the usual four-point alternating group; intersection in one point reduces to that case by conjugation. Explicitly, for triples $\{1, 2, 3\}$ and $\{3, 4, 5\}$, conjugating the second cycle by the first supplies a triple $\{1, 4, 5\}$; these last two triples overlap in two points, and their four-point alternating group joins the first triple in two points. Repeatedly adjoining a triple to a connected union now gives the alternating group on the full component. This proves the assertion. □

Corollary 5 (The internal factor at the head)**.** *Suppose $r \geq 7$ and the original marking has the intersection property. Then $\beta = 1$ in (2). In particular G and its actual tail K have the same parity: G consists entirely of even permutations if and only if K does.*

Proof. Apply Theorem 3 to the tail, whose rank $r-2$ is at least five. In its notation,

$$h_2 = (z\ p)(q\ w),$$

and $\langle h_3, \ldots, h_{r-1}\rangle$ has orbits $\{z\}$, $\{p, q\}$, and one further orbit containing c. The factor β fixes z, c and centralizes this omission group. Its commutation follows from that of h_1 with each h_i, $i \geq 3$: those marks fix x, u, v, z, so they also commute with the other factor of h_1. The three omission orbits have distinct sizes, and fixation of c makes β the identity on the large orbit. Therefore β is either 1 or $(p\ q)$.

Suppose $\beta = (p\ q)$. On the five points b, x, u, v, z the product h_0h_1 is a five-cycle; its remaining factor is β. Thus $(h_0h_1)^5 = \beta$. Conjugating β by h_2 gives $(z\ w)$, so

$$\delta = (p\ q)(z\ w) \in \langle h_0, h_1, h_2\rangle.$$

Lemma 4 applied to the tail puts the same even permutation δ in K. The original intersection property then gives $\delta \in \langle h_2\rangle$, which is false: it is neither the identity nor $(z\ p)(q\ w)$. Hence $\beta = 1$. Both marks in (2) are now even, proving the parity statement. □

Attribution and remaining boundary. The paired selection and ambient orbit-deficit framework comes from the cited Cameron–Fernandes–Leemans–Mixer sources, with the complete recovered proofs and disclosed contributions acquired here. Anchor deletion, the descent excluding two first anchors, and their combination into (2) are the present deduction. The elementary three-cycle argument is standard and has been included to make its use explicit. This component does not acquire the source's near-half classification. The ranks below five, and the internal factor at rank six, still need their exact boundary arguments before excluding a symmetric middle uniformly. No whole-original, conjecture, Lean or measurement credit is claimed.